

04_Class_Activity

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In class activity 4:

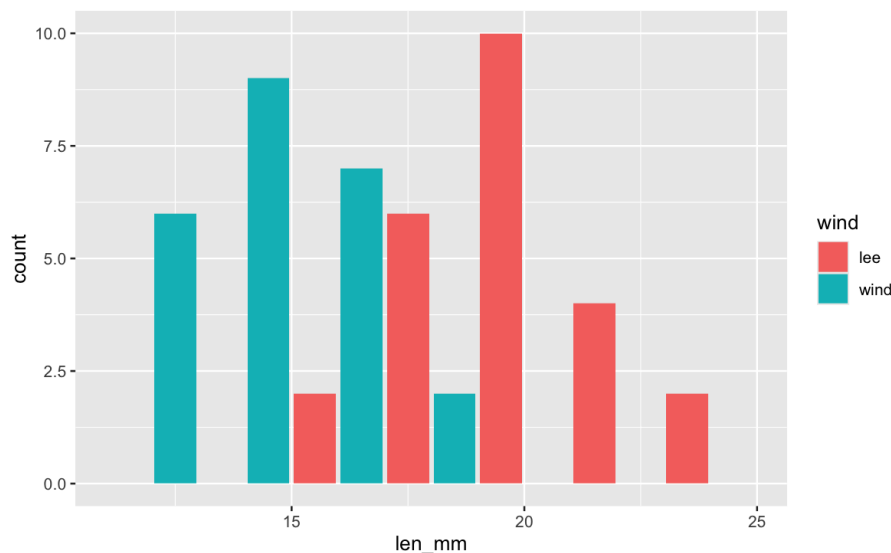
What did we do last time in activity 3?

- Setting up a project and variable names and code names
- How to use the pipe command %>%
- How to create descriptive statistics of a sample

```
p_df %>%  
  summarize(  
    mean_length = mean(length_mm, na.rm = TRUE),  
    sd_length = sd(length_mm, na.rm = TRUE),  
    n_length = sum(!is.na(length_mm)))
```

- More graphs...

```
ggplot(data = p_df, aes(x=length_mm, fill = wind)) +  
  geom_histogram( binwidth = 2,  
  # sets the width in units of the bins - try different nubmers  
  position = position_dodge2(width = 0.5))
```



- What questions do you have and what is unclear - what did not work so far when you started the homework?

Introduction

In this active learning module, we'll explore real data from fish populations in Alaska. We'll focus on understanding:

- How to create and interpret frequency distributions
- How sample size affects our view of a population
- How distributions differ among lakes

We'll use the tidyverse package for data manipulation and visualization.

Setup

First, let's load the packages we need and the dataset:

```
# # Install the patchwork package if needed

# install.packages("patchwork")
library(patchwork)
library(skimr)
library(tidyverse)

# Read in the data file
g_df <- read_csv("data/gray_I3_I8.csv")

i3_df <- g_df %>% filter(lake == "I3")

# Look at the first few rows
head(g_df)
```

```
# A tibble: 6 × 5
  site lake species      length_mm mass_g
<dbl> <chr> <chr>          <dbl>   <dbl>
1  113 I3   arctic grayling    266    135
2  113 I3   arctic grayling    290    185
3  113 I3   arctic grayling    262    145
4  113 I3   arctic grayling    275    160
5  113 I3   arctic grayling    240    105
6  113 I3   arctic grayling    265    145
```

Part 1: Summary Statistics - descriptive statistics

```
# Calculate mean, standard deviation, and sample size by lake
stats_df <- g_df %>%
  group_by(lake) %>%
  summarize(
    mean_length = mean(length_mm, na.rm = TRUE),
    sd_length = sd(length_mm, na.rm = TRUE),
    se_length = sd(length_mm, na.rm = TRUE) / sum(!is.na(length_mm))^.5,
    count = sum(!is.na(length_mm)),
    .groups = "drop"
  ) %>%
  arrange(desc(count))
stats_df
```

```
# A tibble: 2 × 5
  lake mean_length sd_length se_length count
<chr>      <dbl>      <dbl>      <dbl>   <int>
1 I8       363.       52.3       5.18    102
2 I3       266.       28.3       3.48     66
```

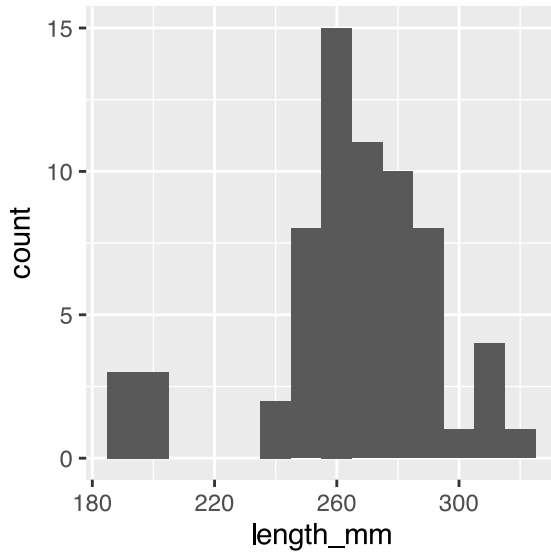
Part 2: Creating Frequency Distributions

Basic Histograms

A histogram shows how many observations fall into certain ranges (or “bins”).

Let’s create a simple histogram of fish lengths from I3 :

```
# Filter for I3 and create a histogram
i3_df %>%
  ggplot(aes(x = length_mm)) +
  geom_histogram(binwidth = 10)
```



💡 Activity 1

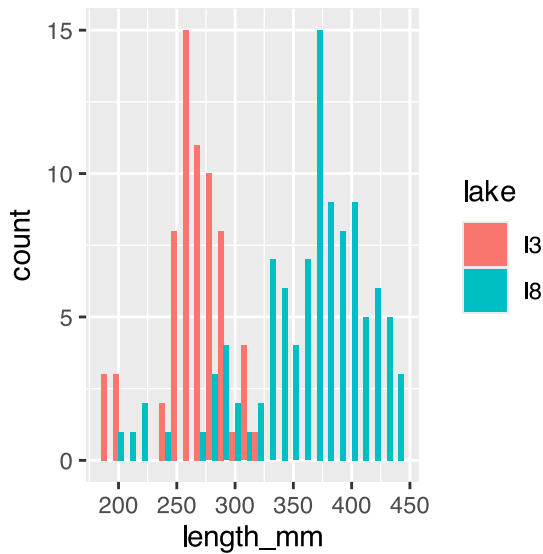
Try changing the binwidth parameter to 5 and then to 1. How does the appearance of the histogram change?

```
# Try it here or above...
```

Comparing Lakes

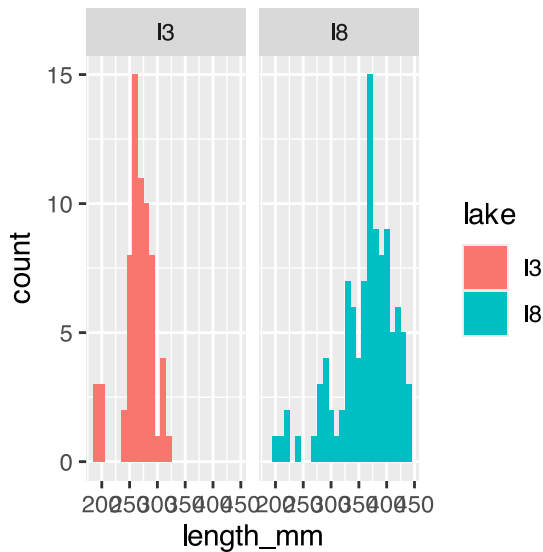
Now let’s compare two lakes

```
# Compare histograms from I3 I8 lakes
g_df %>%
  ggplot(aes(x = length_mm, fill = lake)) +
  geom_histogram(binwidth = 10, position = position_dodge2(width = 0.9))
```



Now let's compare two lakes side by side:

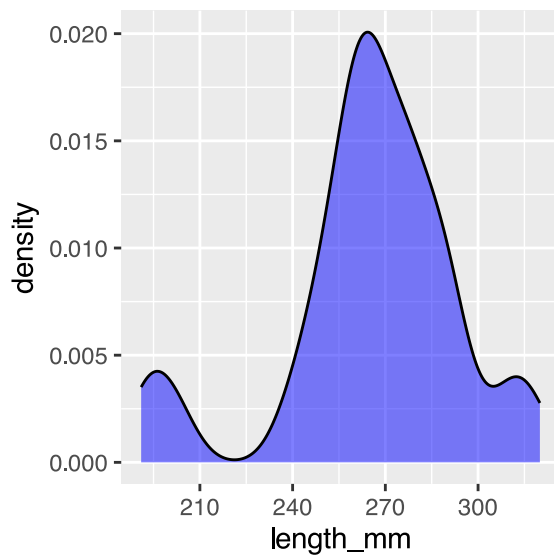
```
# Compare histograms from lake I3 I8
g_df %>%
  ggplot(aes(x = length_mm, fill = lake)) +
  geom_histogram(binwidth = 10) +
  facet_wrap("lake")
```



Part 2: From Histograms to Density Plots

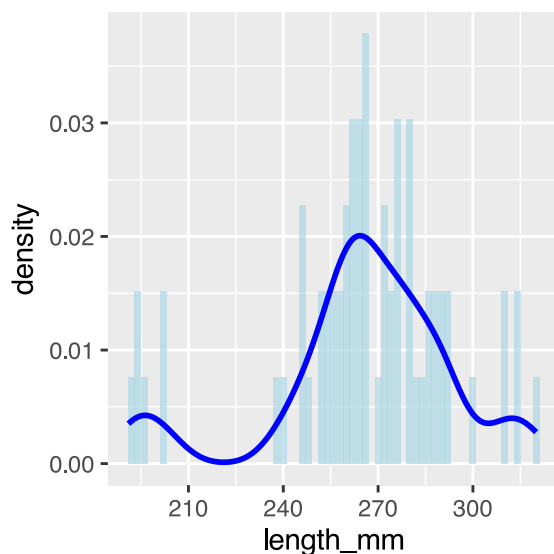
Density plots give us a smoothed version of the histogram. It has the proportion of the data under each part of the curve. This sums to 1.

```
# Create a density plot
i3_df %>%
  ggplot(aes(x = length_mm)) +
  geom_density(fill = "blue", alpha = 0.5)
```



We can overlay the density plot on the histogram :

```
# Combine histogram and density plot
i3_df %>%
  ggplot(aes(x = length_mm)) +
  geom_histogram(aes(y = after_stat(density)), binwidth = 2,
    fill = "lightblue", alpha = 0.7) +
  geom_density(color = "blue", linewidth = 1)
```



Part 3 - area under the density curve

We could show this if we really wanted...

```
# Function to calculate area under density curve
calculate_density_area <- function(data_vector) {
  # Remove NA values
  data_vector <- data_vector[!is.na(data_vector)]

  # Calculate density
  dens <- density(data_vector)
```

```

# Calculate area using numeric integration (trapezoidal rule)
# Area should be approximately 1
dx <- diff(dens$x)
y_avg <- (dens$y[-1] + dens$y[-length(dens$y)]) / 2
area <- sum(dx * y_avg)
return(area)
}

# Apply to i3 lake data
i3_data <- i3_df %>%
  pull(length_mm)

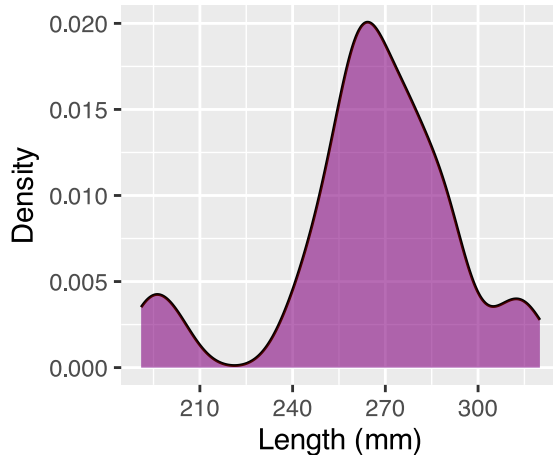
area_value <- calculate_density_area(i3_data)

# Create plot with calculated area
i3_df %>%
  ggplot(aes(x = length_mm)) +
  geom_density(fill = "blue", alpha = 0.4) +
  geom_area(stat = "density", fill = "red", alpha = 0.3) +
  labs(title = "Area Under Probability Density Function = 1",
       subtitle = paste("Calculated area =", round(area_value, 4)),
       x = "Length (mm)",
       y = "Density")

```

Area Under Probability Density

Calculated area = 0.9999



looking at particular areas...

This can be adapted to calculate the area of a subset of the plot

I don't expect you to know or be able to do all of this but is here to play with the code

```

# ----- PART 3: SET INPUT VALUES -----
# change these values to calculate different probabilities
# For this example, let's calculate the probability of fish between 40mm and 60mm
lower_bound <- 320 # change this value
upper_bound <- 350 # change this value

```

```

# ----- PART 1: PREPARE THE DATA -----
# Filter data for just one lake to keep it simple for students
i3_fish <- i3_df %>%
  filter(!is.na(length_mm)) # Remove any missing values

# ----- PART 2: CREATE A FUNCTION TO CALCULATE PROBABILITY -----
# This function calculates the probability of finding a fish with length between
# lower_bound and upper_bound using the empirical distribution of our data
calculate_probability <- function(data_vector, lower_bound, upper_bound) {
  # First, we create a density object from our data
  dens <- density(data_vector)

  # Find indices of x-values that fall within our bounds
  indices <- which(dens$x >= lower_bound & dens$x <= upper_bound)

  # If we have no points in the range, return 0
  if(length(indices) <= 1) {
    return(0)
  }

  # Get x and y values within our bounds
  x_values <- dens$x[indices]
  y_values <- dens$y[indices]

  # Calculate the area using the trapezoidal rule
  # (average height × width) for each segment, then sum all segments
  widths <- diff(x_values)
  avg_heights <- (y_values[-1] + y_values[-length(y_values)]) / 2
  area_in_range <- sum(widths * avg_heights)

  # Return the calculated probability
  return(area_in_range)
}

# ----- PART 4: CALCULATE THE PROBABILITY -----
# Calculate the probability for the specified range
probability <- calculate_probability(i3_fish$length_mm, lower_bound, upper_bound)

# Calculate the total area to show that the complete distribution sums to approximately 1
total_area <- calculate_probability(i3_fish$length_mm,
                                   min(i3_fish$length_mm),
                                   max(i3_fish$length_mm))

# ----- PART 5: CREATE THE VISUALIZATION -----
# Create density data for the highlighting
density_data <- density(i3_fish$length_mm)
density_df <- data.frame(x = density_data$x, y = density_data$y)

# Create a subset for the area of interest
highlight_df <- density_df %>%
  filter(x >= lower_bound & x <= upper_bound)

```

```

# Create the plot
ggplot(i3_fish, aes(x = length_mm)) +
  # First, plot the overall density curve in light blue
  geom_density(fill = "lightblue", alpha = 0.5) +

  # Then highlight our region of interest in dark red
  geom_area(data = highlight_df, aes(x = x, y = y),
           fill = "darkred", alpha = 0.7) +

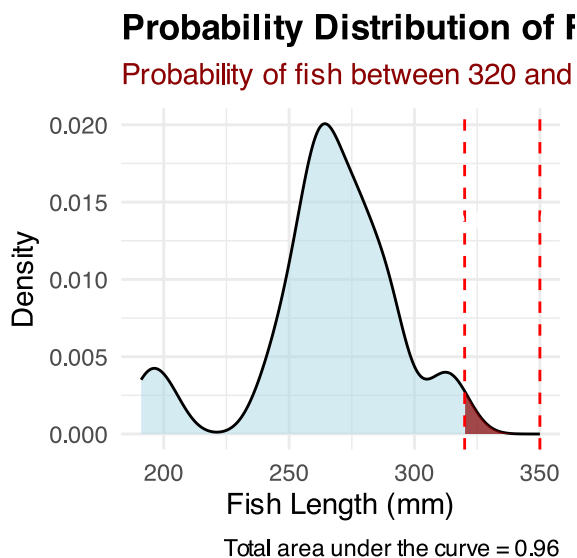
  # Add vertical lines to clearly mark the boundaries
  geom_vline(xintercept = lower_bound, linetype = "dashed", color = "red") +
  geom_vline(xintercept = upper_bound, linetype = "dashed", color = "red") +

  # Add informative labels
  labs(
    title = "Probability Distribution of Fish Lengths",
    subtitle = paste0("Probability of fish between ", lower_bound,
                      " and ", upper_bound, " mm = ",
                      round(probability * 100, 1), "%"),
    caption = paste("Total area under the curve =", round(total_area, 3)),
    x = "Fish Length (mm)",
    y = "Density"
  ) +

  # Add text annotations to explain the areas
  annotate("text", x = (lower_bound + upper_bound)/2,
          y = max(density(i3_fish$length_mm)$y) * 0.7,
          label = paste0("Area = ", round(probability, 3)),
          color = "white", size = 4) +

  # Make the plot look nicer
  theme_minimal() +
  theme(
    plot.title = element_text(face = "bold"),
    plot.subtitle = element_text(color = "darkred")
  )

```



Part 4: this is great but integrating area each time is a pain

Converting data to Z scores

```
# Calculate the mean and standard deviation of fish lengths
mean_length <- mean(i3_df$length_mm, na.rm = TRUE)
sd_length <- sd(i3_df$length_mm, na.rm = TRUE)

# Calculate Z-scores for fish lengths
i3_df <- i3_df %>%
  mutate(z_score = (length_mm - mean_length) / sd_length)

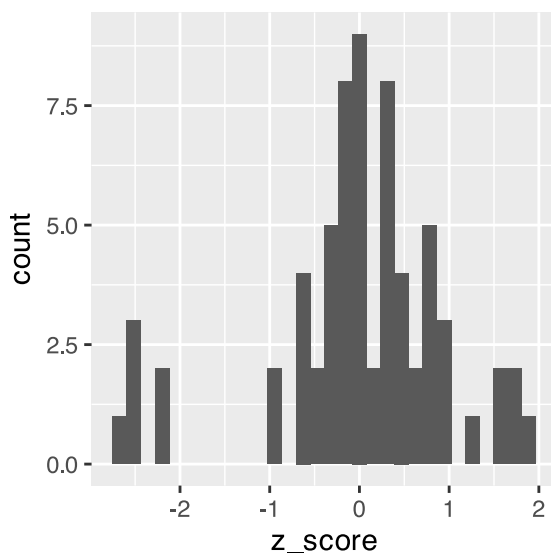
# View the first few rows with Z-scores
head(i3_df)
```

```
# A tibble: 6 × 6
  site lake species length_mm mass_g z_score
<dbl> <chr> <chr>      <dbl>   <dbl>   <dbl>
1  113 I3  arctic grayling    266    135  0.0139
2  113 I3  arctic grayling    290    185  0.862
3  113 I3  arctic grayling    262    145 -0.127
4  113 I3  arctic grayling    275    160  0.332
5  113 I3  arctic grayling    240    105 -0.905
6  113 I3  arctic grayling    265    145 -0.0214
```

Now plot the Z Scores as a histogram

```
z_fish_plot <- i3_df %>%
  ggplot(aes(x=z_score)) +
  geom_histogram()
z_fish_plot
```

`stat\_bin()` using `bins = 30`. Pick better value with `binwidth`.



we can use this now to get the area the same way as above but easier...

Proportion within 1 standard deviation = sum of absolute values of Z Scores that are less than or equal to 1 divided by the number in the sample...

Remember in a true normal distribution it is 68% within 1 std dev.

should be approximately (varies if distribution is not normal):

- 68% of data within $\pm 1\sigma$ of the mean
- 95% of data within $\pm 2\sigma$ of the mean - really 1.96σ
- 99.7% of data within $\pm 3\sigma$ of the mean

```
# What proportion of fish are within 1 standard deviation of the mean?
within_lsd <- sum(abs(i3_df$z_score) <= 1, na.rm = TRUE) / sum(!is.na(i3_df$z_score))
cat("Proportion within 1 SD:", round(within_lsd * 100, 1), "%\n")
```

```
Proportion within 1 SD: 81.8 %
```

Z-score example calculation in r

We can use R to get these values easier...

For standard normal distribution (mean=0, sd=1):

- `pnorm(z)` # gives cumulative probability (area to the left)
- `qnorm(p)` # gives z-value for a given probability
- `dnorm(z)` # gives probability density

```
# Examples:
z_value <- 1.22
prob_left <- pnorm(z_value)           # 0.975 (97.5% to the left)
prob_right <- 1 - pnorm(z_value)      # 0.025 (2.5% to the right)
prob_between <- pnorm(2) - pnorm(-2) # 0.95 (95% between  $\pm 1.96$ )

# To find z-value for a given probability:
z_for_95_percent <- qnorm(0.888)      # 1.96

print(prob_left)
```

```
[1] 0.8887676
```

```
print(prob_right)
```

```
[1] 0.1112324
```

```
print(prob_between)
```

```
[1] 0.9544997
```

```
print(z_for_95_percent)
```

```
[1] 1.21596
```

We can now use this for fun in the fish

Lets say we are interested in knowing at what point from I3 it is not likely to catch a larger fish?

Maybe we expect 95% of the time to catch a fish that is “common” but the 5% is the unlikely portion....

```
# Examples:
# What fish length corresponds to the top 5% (unlikely)?
top_5_percent_z <- qnorm(0.95) # z-score for 95th percentile
unlikely_length <- mean_length + (top_5_percent_z * sd_length)

cat("Only 5% of fish are longer than:", round(unlikely_length, 1), "mm\n")
```

```
Only 5% of fish are longer than: 312.2 mm
```

```
cat("This corresponds to z-score:", round(top_5_percent_z, 3), "\n")
```

```
This corresponds to z-score: 1.645
```

Part 5: Comparing a sampe mean to an expected mean....

Did the smaple come from that lake?

Lets practice a One-Sample t-Test

Let’s perform a one-sample t-test to determine if the mean fish length in Lake I3 differs from 260 mm:

```
# get only lake I3
i3_df <- g_df %>% filter(lake=="I3")

# what is the mean
i3_mean <- mean(i3_df$length_mm, na.rm=TRUE)
cat("Mean:", round(i3_mean, 1), "mm\n")
```

```
Mean: 265.6 mm
```

```
# Perform a one-sample t-test
t_test_result <- t.test(i3_df$length_mm, mu = 260)

# View the test results
t_test_result
```

One Sample t-test

```
data: i3_df$length_mm
t = 1.6091, df = 65, p-value = 0.1124
alternative hypothesis: true mean is not equal to 260
95 percent confidence interval:
 258.6481 272.5640
```

```
sample estimates:
mean of x
265.6061
```

Part 6: Comparing two means

Formulating Hypotheses

For the following research questions about Arctic grayling, write the null and alternative hypotheses:

1. Are fish in Lake I8 longer than fish in Lake I3?

```
# Let's test one of these hypotheses: Are fish in Lake I8 longer than fish in Lake I3?

# Perform an independent t-test
t_test_result <- t.test(length_mm ~ lake, data = g_df,
                        alternative = "less") # H0:  $\mu_{I3} \geq \mu_{I8}$ , H1:  $\mu_{I3} < \mu_{I8}$ 

# Display the results
t_test_result
```

Welch Two Sample t-test

```
data: length_mm by lake
t = -15.532, df = 161.63, p-value < 2.2e-16
alternative hypothesis: true difference in means between group I3 and group I8 is less than 0
95 percent confidence interval:
 -Inf -86.66138
sample estimates:
mean in group I3 mean in group I8
    265.6061      362.5980
```